

Practice 5

A. Use guess-and-check to solve the following problems.

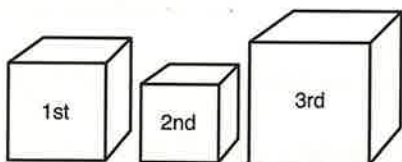
1. A number divided by 2 is equal to 12 less than the original number. What is the number?

A. 12
B. 20
C. 24
D. 28

2. For a fundraiser, Sandra raised three times as much money as Barbara, and Barbara raised \$50 more than Matt. Together they raised \$950. How much money did Barbara raise?

A. \$150
B. \$175
C. \$200
D. \$325

3. The three packages below weigh a total of 15 pounds.



The first package weighs twice as much as the second package. The third package weighs three times as much as the second package. How many pounds does the first package weigh?

A. 2
B. 4
C. 5
D. $6\frac{1}{2}$

4. Hannah scored a total of 170 points on two math tests. The score of the first test was 6 points lower than the score of the second test. How many points did Hannah score on the first test?

A. 76
B. 82
C. 88
D. 90

5. Nelson is twice as old as Maria. Six years ago, Nelson was five times as old as Maria. How old was Nelson six years ago?

A. 5
B. 10
C. 15
D. 20

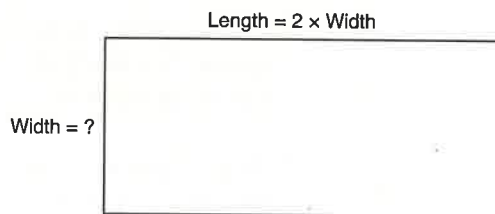
6. Which of the following is a solution for the quadratic equation $2x^2 + x - 15 = 0$?

A. -3
B. -1
C. 2
D. 3

7. An amusement park sells adults' and children's passes. An adult's pass is \$25, and a child's pass is \$15. A group spent \$440 on 20 passes. How many children's passes did the group purchase?

A. 5
B. 6
C. 9
D. 14

8. The rectangular garden below is twice as long as it is wide. If the total distance around the garden is 120 feet, what is the width of the garden in feet?



A. 15
B. 20
C. 30
D. 40

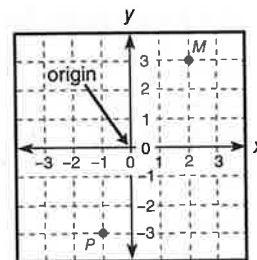
Answers and explanations begin on page 677.

Lesson 6: The Coordinate Plane

Key Ideas

- A coordinate grid is formed by two intersecting axes, or number lines.
- The x -axis is horizontal, and the y -axis is vertical.
- The location of a point is shown by two numbers called an ordered pair: (x, y) .

A **coordinate grid** is a way to locate points that lie in a **plane**, or flat surface. The grid is formed by two intersecting lines, an x -axis and a y -axis. The x -axis is actually a horizontal number line, and the y -axis is a vertical number line. The point at which the two axes intersect is called the **origin**.

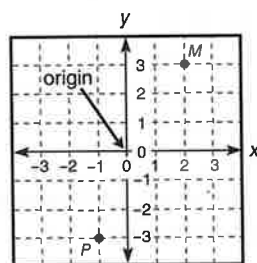


Each point on the grid can be named using two numbers called an **ordered pair**. The first number is the distance from the origin along the x -axis. The second number is the distance from the origin along the y -axis.

The numbers are written in parentheses and are separated by a comma: (x, y) .

Example 1: Write the ordered pairs for points M and P .

1. Point M lies 2 spaces to the right of the origin along the x -axis and 3 spaces above the origin along the y -axis. The coordinates are $(2, 3)$.
2. Point P lies 1 space to the left along the x -axis and 3 spaces down along the y -axis. The coordinates are $(-1, -3)$.



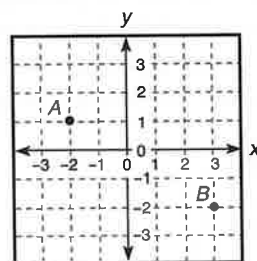
GED® TEST TIP

If you are given a coordinate grid that is not labeled, remember that the origin is at $(0,0)$. Starting at the origin, count either right or left first, then up or down.

To plot points on the grid, use the number lines located at the axes. Remember that right and up are the directions for positive numbers and left and down are the directions for negative numbers.

Example 2: Point A is located at $(-2, 1)$, and point B is located at $(3, -2)$. Plot these points on a coordinate grid.

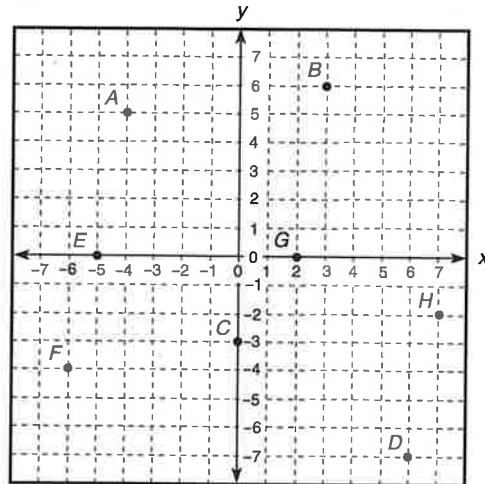
1. To plot point A , start at the origin. Count 2 spaces left along the x -axis. Count 1 space up along the y -axis.
2. To plot point B , start at the origin. Count 3 spaces right along the x -axis. Count 2 spaces down along the y -axis.



Practice 6

A. Write the ordered pair for each point.

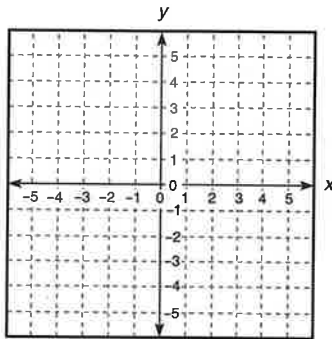
1. Point A
2. Point B
3. Point C
4. Point D
5. Point E
6. Point F
7. Point G
8. Point H



B. Plot the points on the coordinate grid.

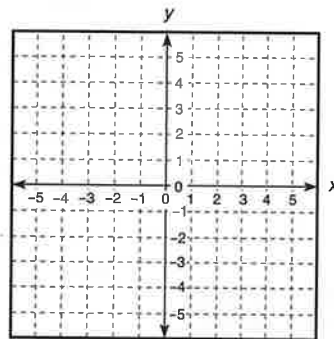
9. Plot the following points:

J at $(-3, -2)$
 K at $(4, 0)$
 L at $(1, -3)$
 M at $(-4, 2)$



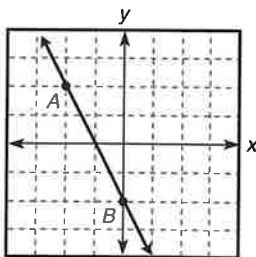
10. Plot the following points:

N at $(0, -1)$
 O at $(-4, -4)$
 P at $(3, 1)$
 Q at $(-3, 0)$



C. Choose the one best answer to each question.

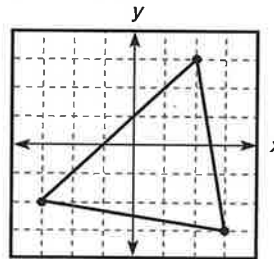
11. On the coordinate grid below, a line passes through points A and B.



Which of the following ordered pairs also lies on the line?

- A. $(1, 0)$
- B. $(1, -1)$
- C. $(0, -1)$
- D. $(-1, 0)$

12. Two of the corners of a triangle are located at $(3, -3)$ and $(2, 3)$. What is the location of the third corner as shown in the diagram below?



- A. $(-3, -2)$
- B. $(-3, 2)$
- C. $(-2, -3)$
- D. $(3, -2)$

Answers and explanations begin on page 678.

Lesson 7: Graphing a Line

Using the coordinate system, we can graph equations. When an equation has only two variables, x and y , and neither is raised to an exponent, the graph of the equation will be a line. When the graph of an equation is a straight line, the equation is a **linear equation**.

Key Ideas

- A linear equation has two variables, x and y .
- When the solutions to a linear equation are graphed on a coordinate grid, the graph forms a line.
- To find a point on the line, substitute a value for x and solve for y .
- You must solve for at least two points in order to draw the line.

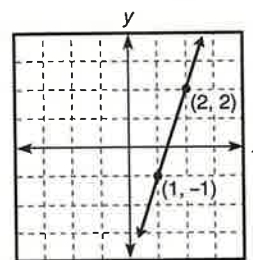
To graph an equation, you need to solve for two points on the line.

Example 1: Graph the equation $y = 3x - 4$.

- | | | |
|---|---|---|
| 1. Choose any value for x and solve for y . Let $x = 1$. | $y = 3(1) - 4$
$y = 3 - 4$
$y = -1$ | If $x = 1$, then $y = -1$.
The ordered pair for the first point is $(1, -1)$. |
| 2. Choose another value for x and solve for y . Let $x = 2$. | $y = 3(2) - 4$
$y = 6 - 4$
$y = 2$ | If $x = 2$, then $y = 2$. The ordered pair for the second point is $(2, 2)$. |

3. Plot the points on a coordinate grid and draw a line through them.

The line is the graph of all the possible solutions for the equation $y = 3x - 4$. Arrows at both ends of the line indicate that the line continues in both directions. From this, you can see that there is an infinite number of solutions to a linear equation.



GED® TEST TIP

If a linear equation is not written with y on one side of the equation, use inverse operations to isolate y .
Example: $2x + y = 15$.
Subtract $2x$ from each side.
 $y = -2x + 15$.

Some linear equation problems don't require you to draw a graph.

Example 2: Point A lies at $(5, -6)$ on a coordinate grid. The graph of which of the following equations passes through point A?

- (1) $y = -5x + 18$
- (2) $y = -4x + 14$
- (3) $y = -2x - 13$

Use the ordered pair given in the problem. Substitute the x -coordinate, 5, for x in each equation and solve for y . If $y = -6$, the value of the y -coordinate from the ordered pair, you have found the correct equation.

Option (2) is correct.

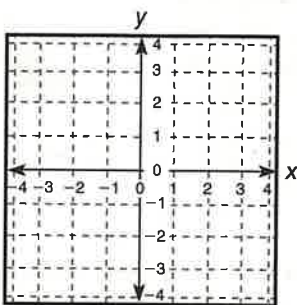
$$y = -4x + 14$$
$$y = -4(5) + 14$$
$$y = -20 + 14 = -6$$

Practice 7

A. Fill in the y column in each table and graph the equation.

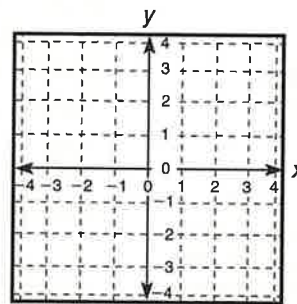
1. $y = \frac{1}{2}x + 3$

If $x =$	then $y =$
-2	
0	
2	



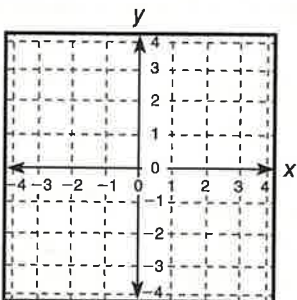
3. $-2 + y = -x$

If $x =$	then $y =$
1	
2	
3	



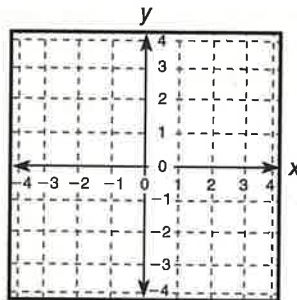
2. $y + 3x = -1$

If $x =$	then $y =$
-1	
0	
1	



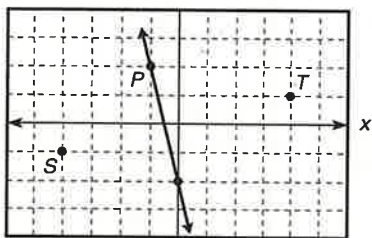
4. $y = 3 - 2x$

If $x =$	then $y =$
0	
1	
2	



B. Choose the one best answer to each question.

Questions 5 and 6 refer to the following coordinate grid.



5. The graph of the equation $y = \frac{1}{4}x$ will pass through which of the following pairs of points?

- A. point S and $(-1, 2)$
- B. point S and $(0, -2)$
- C. point T and $(0, 0)$
- D. point T and $(0, -2)$

6. Line P is the graph of which of the following equations?

- A. $y = 4x + 1$
- B. $y = -4x - 1$
- C. $y = 4x + 2$
- D. $y = -4x - 2$

7. Point C is located at $(-3, 5)$. A graph of which of the following equations would pass through point C?

- A. $3x + 2y = 5$
- B. $2x + 3y = 9$
- C. $4x - 2y = 8$
- D. $3x - 3y = 6$

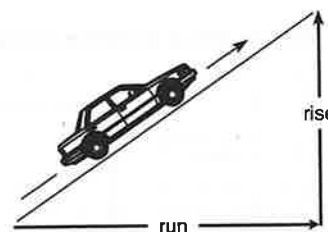
Answers and explanations begin on page 678.

Lesson 8: Slope of a Line

Key Ideas

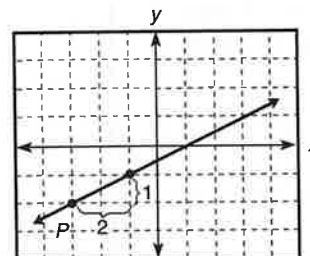
- Slope is the ratio of rise to run.
- Moving from left to right, a line that goes upward has a positive slope, and a line that moves downward has a negative slope.
- You can find slope by counting spaces and writing a ratio or by using the slope formula.

Slope is the measurement of the steepness of a line. Imagine a road going up a hill. If the road must climb upward over a short forward distance, the road will be very steep. Slope measures the relationship between **rise** (how high the road must climb) and **run** (the distance the road goes forward).



On a coordinate grid, a line that moves upward from left to right has a **positive slope**. A line that moves downward from left to right has a **negative slope**. You can find the slope of a line on a coordinate grid by writing the ratio of rise to run.

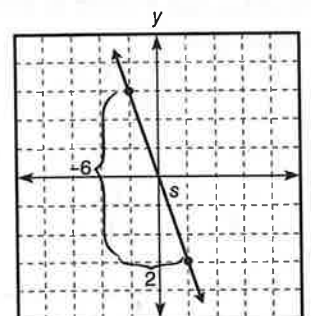
Example 1: What is the slope of line P shown on the coordinate grid?



1. Find two points on line P . Count to find the rise and run. The line moves up 1 space for every 2 spaces it goes to the right.

2. Write the ratio: $\frac{\text{rise}}{\text{run}} = \frac{1}{2}$. The slope is $\frac{1}{2}$.

Example 2: What is the slope of line s shown on the coordinate grid?



1. Find any two points on line s . The line moves down 6 spaces (a negative direction) and 2 spaces to the right.

2. Write the ratio: $\frac{\text{rise}}{\text{run}} = \frac{-6}{2} = -3$.

The slope of line s is -3 .

You can also find slope using the slope formula on the GED® Formula Sheet. The formula will appear as follows:

slope of a line (m) = $\frac{y_2 - y_1}{x_2 - x_1}$, where (x_1, y_1) and (x_2, y_2) are two points on a line.

Example 3: A line passes through points at coordinates $(1, 4)$ and $(-5, 2)$. What is the slope of the line?

1. Choose one point to be (x_1, y_1) . The other will be (x_2, y_2) . It doesn't matter which you choose. For this example, $(x_1, y_1) = (1, 4)$ and $(x_2, y_2) = (-5, 2)$.

GED® TEST TIP

Think of the slope formula this way: write the difference of the y's over the difference of the x's. Just make sure that you always subtract the coordinates in the same order.

2. Substitute the values into the slope formula and solve:

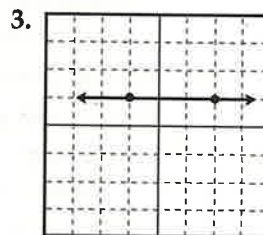
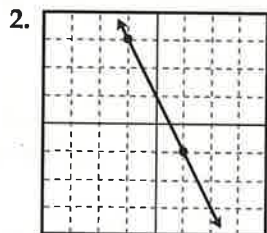
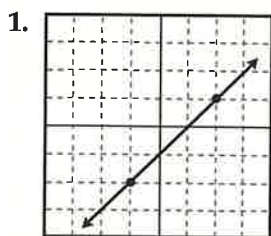
$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - 4}{-5 - 1} = \frac{-2}{-6} = \frac{1}{3} \quad \text{The slope is } \frac{1}{3}.$$

Since the slope is positive, you know that the line rises from left to right. You also know that it goes up 1 space for every 3 spaces it moves to the right.

In working with slope, there are a few special circumstances that you should memorize. A horizontal line, just like a flat stretch of roadway, has a **slope of 0**. The slope of a vertical line is **undefined**; in other words, our definition of slope will not work for a line that has no run at all.

Practice 8

A. Find the slope of each line.



B. Use the slope formula to find the slope of a line that passes through the following pairs of points.

4. (3, 5) and (-1, 2)

6. (4, 2) and (2, 2)

8. (1, 4) and (-2, -2)

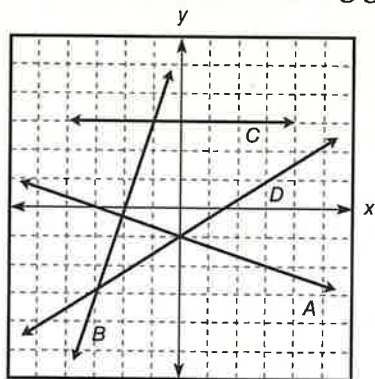
5. (0, 2) and (4, 0)

7. (6, 1) and (0, 3)

9. (4, -2) and (2, 4)

C. Choose the one best answer for each question.

Question 10 refers to the following graph.



10. Which of the following lines shown on the graph has a slope of $-\frac{1}{3}$?

- A. line A
- B. line B
- C. line C
- D. line D

11. Line N passes through the following points: (0, 4), (1, 2), (2, 0), and (3, -2). What is the slope of line N?

- A. -4
- B. -2
- C. 2
- D. 4

12. Line L passes through point (1, 0) and has a slope of 3. Which of the following points also lies on line L?

- A. (0, 3)
- B. (1, 3)
- C. (2, 3)
- D. (2, 5)

Answers and explanations begin on page 678.

Lesson 9: Slope and Equations

Use the Slope-Intercept Form to Find an Equation

Key Ideas

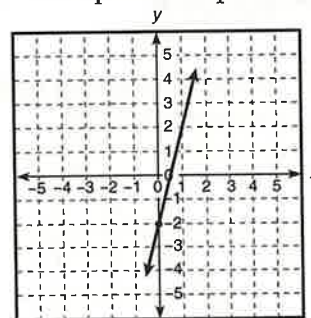
- There are two ways to find an equation if you are given the slope of a line and a point.
- One method is the slope-intercept form.
- The other method is the point-slope form.

Lesson 8 showed how to find the slope of a line and graph it. The examples in this lesson show how you can use the slope and a point on the line to find the equation of a line in two different forms. (You might also need to find the equation of a line from two points. In this case, you would calculate the slope first, then use the slope and one of the points to find the equation.)

The first of these forms is the **slope-intercept form**: $y = mx + b$. In this form of the equation, the variable m stands for the **slope** of the line. The variable b stands for the **y-intercept**, which is the y -value at the point where the line crosses the y -axis. The variables x and y are the x - and y -coordinates of any point on the line and are usually written as an ordered pair.

Follow these steps to find the equation of a line in the slope-intercept form.

1. Substitute the values that you are given for the slope (m) and the x - and y -coordinates (x, y) into the slope-intercept equation. Be careful not to mix up x and y .
2. Use inverse operations to isolate b .
3. Rewrite the equation in slope-intercept form, leaving x and y as variables and substituting values for m and b .



Example 1: Use the slope-intercept form to find the equation of a line that has the slope $m = 4$ and passes through the point $(-1, -6)$.

1. $-6 = (4) \times (-1) + b$
 $-6 = -4 + b$
2. $-2 = b$
3. $y = 4x - 2$

Use the Point-Slope Form to Find an Equation

The second important form that is used to describe a line on the *Mathematical Reasoning Test* is the **point-slope form**: $y - y_1 = m(x - x_1)$. In this form, (x_1, y_1) is an ordered pair that corresponds to a point on the line. As in the slope-intercept form, m stands for the slope. If you simplify an equation in point-slope form by solving for y , you will get the slope-intercept form of the equation.

Follow these steps to find the equation of a line in the point-slope form.

1. Call the point that you are given (x_1, y_1) .
2. Put the x_1 and y_1 values into the point-slope equation.
3. Put the slope value in the point-slope equation for m .

GED® TEST TIP

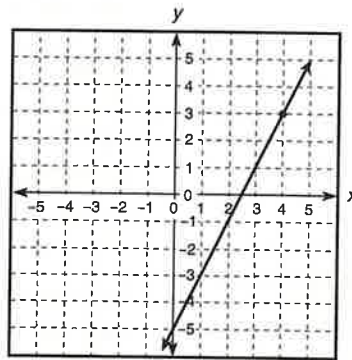
You can access both the slope-intercept form and the point-slope form from the online GED® Formula Sheet. Just click on the word formula on the computer toolbar, and the sheet will become available.

Example 2: Use the point-slope form to find the equation of the line that passes through the point $(4, 3)$ and has a slope of 2.

1. $(x_1, y_1) = (4, 3)$
2. $y - 3 = m(x - 4)$
3. $y - 3 = 2(x - 4)$

Notice that you can simplify the equation to find the slope-intercept form:

$$\begin{aligned} y - 3 &= 2(x - 4) \\ y - 3 &= 2x - 8 \\ y &= 2x - 5 \end{aligned}$$



Practice 9.1

A. Use the slope-intercept form, $y = mx + b$, to find the equation of the line that passes through the given point and has the given slope.

1. $(1, -2); m = -4$
2. $(-1, -4); m = 2$
3. $(-4, 2); m = -\frac{1}{3}$

B. Use the point-slope form, $y - y_1 = m(x - x_1)$, to find the equation of the line that passes through the given point and has the given slope.

4. $(2, 1); m = 3$
5. $(2, 0); m = -\frac{1}{3}$
6. $(1, -2); m = 1$

C. Find the equation of the line that passes through the given points. Write your answer in slope-intercept form.

7. $(-5, 3), (1, 1)$
8. $(-3, 0), (-2, 4)$
9. $(-3, -4), (7, 1)$

D. Choose the one best answer for each question.

10. Which of the following is an equation for the line that passes through $(-1, 0)$ and $(2, -3)$?

- A. $y = 3x + 3$
- B. $y = -3x + 9$
- C. $y = -x - 1$
- D. $y = x - 5$

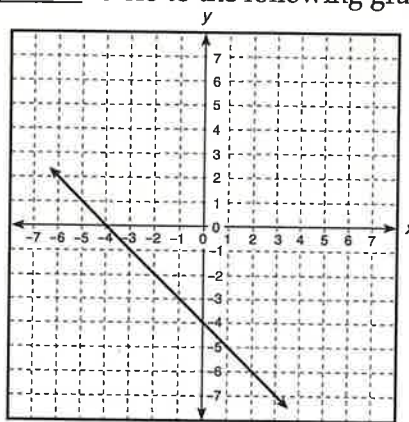
11. Which of the following is an equation for the line that passes through $(-3, 4)$ and $(1, 4)$?

- A. $y = 0$
- B. $y = 4$
- C. $y = x + 3$
- D. $y = x + 7$

12. Which of the following equations describes the same line as $y - 2 = \frac{1}{2}(x - 6)$?

- A. $y = 3x + 2$
- B. $y = \frac{1}{2}x - 4$
- C. $y = -x + 12$
- D. $y = \frac{1}{2}x - 1$

Question 13 refers to the following graph.



13. Which of the following equations correctly describes the line on the graph?

- A. $y = x - 4$
- B. $y = -x - 4$
- C. $y = x + 4$
- D. $y = -x + 4$

Answers and explanations start on page 679.

Key Ideas

- The slope of a line is equivalent to rate when the slope is written as a whole number (e.g., 55 miles per hour) or decimal (e.g., \$2.89 per pound).
- Rate is usually written as number of units of quantity y per *one unit* of quantity x . This value is called the *unit rate*.
- It is possible to determine rate with just one value each for the y and x variables when you know that when the x value is 0, the y value is also 0 (e.g., if you buy zero apples, you pay zero dollars).

GED® TEST TIP

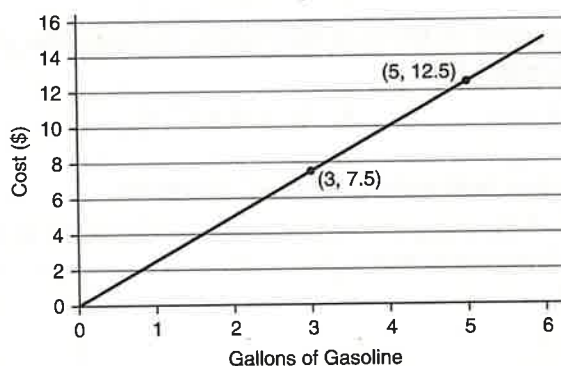
To determine rate, just take the fraction that represents slope and divide the numerator (top number) by the denominator (bottom number). Either version of the slope formula, $\frac{\text{rise}}{\text{run}}$ or $\frac{y_2 - y_1}{x_2 - x_1}$, may be used to determine rate.

Slope and Unit Rate

The **unit rate** is a way of showing the relationship between two variables as the ratio of how many units of one quantity there are *per one unit of the other quantity*. Unit rates are used constantly in everyday life, as well as in business, science, and the trades. On a given day, you might go to the butcher for a cut of meat that costs \$2.89 per pound; merge onto a highway where you increase your speed to 55 miles per hour; and compare LED and fluorescent light bulbs at the hardware store, seeing that the fluorescents use an extra 1.8 kJ of electricity per hour.

Each of these everyday ratios compares two quantities, and the second quantity given (per *this*) is what would be represented by the x -axis of a graph. The unit rate, the quantity of y per one unit of x , is the same as the slope of a line representing the relationship, calculated in the previous lesson as $\frac{\text{rise}}{\text{run}}$, with one difference. The unit rate is typically expressed in whole numbers or decimals rather than left in fraction form.

Example 1: The coordinate grid shows the cost of gas at a particular station. Avery and Beatrix partially filled two jerry cans, separately, and later graphed the values from their receipts to determine the relationship between cost (y) and gallons of gas purchased (x). Determine the cost of gas from the coordinate grid.

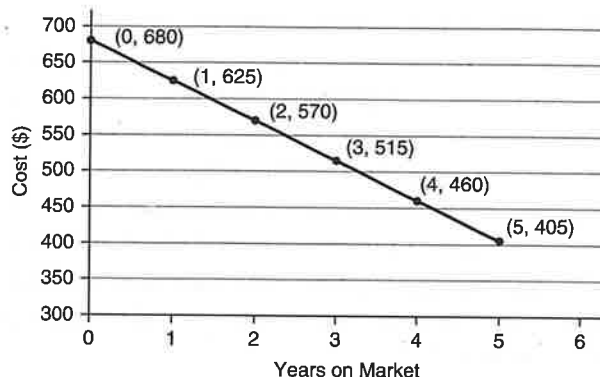


1. Count to find the rise and run between the two given points. The line moves up 5 when it moves 2 to the right.
2. Write the ratio: $\frac{\text{rise}}{\text{run}} = \frac{5}{2} = 2.5$. (This is the same as saying that the ratio is $\frac{2.5}{1}$.)
3. Interpret the meaning of the rate as determined by the slope: the cost of gas at this station is \$2.50 per gallon.

NOTE: You may have noticed that in this example, you can get the same answer by just dividing 7.5 by 3, or 12.5 by 5. This works because there is an implied third point in this proportional relationship: when you buy 0 gallons of gas, the cost is 0 dollars, meaning $\frac{7.5 - 0}{3 - 0} = \frac{7.5}{3} = 2.5$.

Example 2: A technology company follows a predictable pricing trend, where the price of a new product decreases linearly over time. The company releases a new smart phone at a retail price of \$680.

The following graph shows the cost of the phone for the next five years it is on the market as cost (y) versus time in years (x).



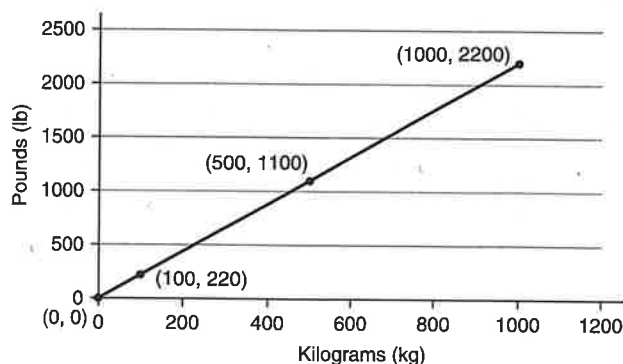
Determine the rate at which the discount in pricing occurs.

1. Choose any two points and use the formula for slope of a line, $(m) = \frac{y_2 - y_1}{x_2 - x_1}$, to find the rate. For this example, you could choose $(x_1, y_1) = (0, 680)$ and $(x_2, y_2) = (5, 405)$.
2. Substitute the values into the slope formula and solve, remembering to reduce to a whole number or decimal rather than leaving the slope as a fraction:

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{405 - 680}{5 - 0} = \frac{-275}{5} = -55$$
3. Interpret the meaning of the rate as determined by the slope. The smart phone model drops in price by \$55 every year it is on the market.

Example 3: A Minnesota construction company orders limestone, sand, and topsoil from a Canadian company, which uses the metric system. The purchasing officer knows that there are 2.2 lb per 1 kg of material. He decides to make a graph for reference when making purchases. Typically he'll buy 100 kg, 500 kg, or 1000 kg of materials at a time. What will the graph look like?

1. The material weighs 2.2 lbs per 1 kg. Use the metric kilogram measurements as x values and the corresponding imperial pound measurements as y values. For 100 kg, $2.2(100) = 220$ lb. For 500 kg, $2.2(500) = 1100$ lb. For 1000 kg, $2.2(1000) = 2200$ lb.
2. Plot the points on a graph.



Note that the slope of the line connecting the dots is 2.2, representing the unit conversion rate of 2.2 lb per kg. The purchasing officer could use the graph or the equation of the line to find the weight in pounds of nonstandard orders.

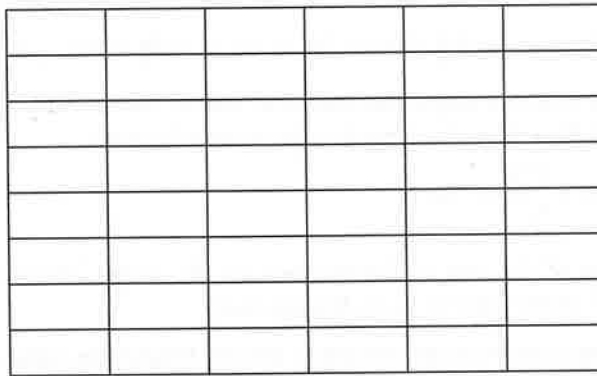
Practice 9.2

A. Use the slope formula to find the rates that correspond to the numbers in these scenarios.

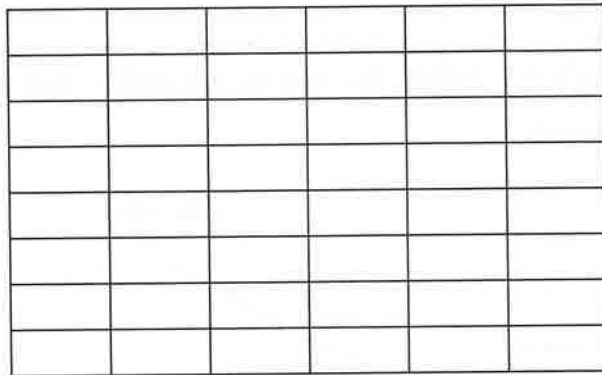
1. A catering company orders 150 lb of fresh clams for a banquet serving 100 people, and 30 lb of clams for a dinner party of 20 people. Determine the standard number of pounds of clams per serving.
2. A painter uses 15 gallons of interior paint to cover 300 square yards of wall and ceiling space. On a different job, he uses 8 gallons to cover 160 square yards. No paint is left over on either job. How many gallons of paint are needed to cover 1 square yard?
3. A scientist is looking for the relationship between the x and y variables in an experiment, and she collects the following data: $(3, 2.4)$, $(8, 6.4)$, $(9, 7.2)$. Determine the rate of quantity y per unit of quantity x .

B. Given the rate, draw a graph showing the relationship between the following quantities.

4. Oranges are on sale for \$0.44 per pound. Draw a graph of cost (y) per weight of oranges (x).

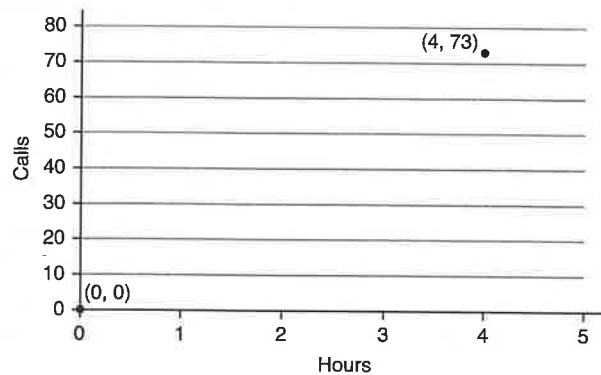


5. A vintage vinyl record player spins at a rate of 78 rpm (revolutions per minute). Draw a graph of number of revolutions (y) per time (x), plotting points for a 3-minute, 4-minute, and 5-minute record.



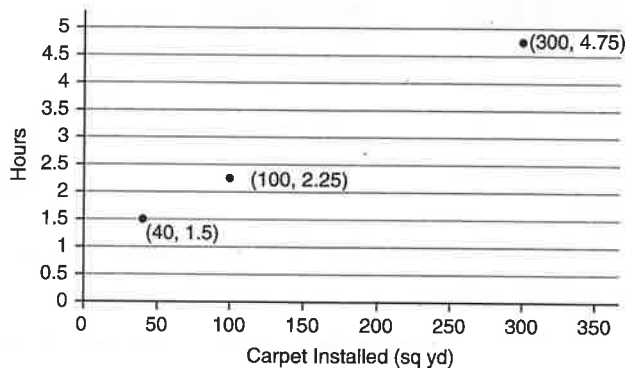
C. Given the following graphs, determine the rates.

Question 6 refers to the following graph.



6. A telemarketing company tracks its employees' efficiency. Use the graph to find the rate of calls made per hour of work.
- A. 0.055 calls per hr
 - B. 17.25 calls per hr
 - C. 18.25 calls per hr
 - D. 69 calls per hr

Question 7 refers to the following graph.



7. On three carpet installation jobs, a contractor records the values above with the goal of analyzing whether her rates are high enough given the time it takes her to install carpet. It takes her 1 hour to collect tools and drive to a worksite. The x -axis shows square yards of carpet, and the y -axis shows hours spent. What is her rate of time per area installed?
- A. 0.0125 hr per sq yd
 - B. 0.01625 hr per sq yd
 - C. 0.0375 hr per sq yd
 - D. 80 hr per sq yd

Answers and explanations begin on page 679.

Lesson 10: Systems of Linear Equations

Graph to Solve the System of Equations

The equations in Lessons 7 and 9 are **linear equations**: they have two variables, x and y , and represent straight lines in the coordinate plane. Two or more linear equations make up a **system of linear equations**. Solving a system of two equations means finding the values of both variables. The **solution** will give the x - and y -coordinates of the point at which the two lines intersect. You can express this solution as an ordered pair: (x, y) .

One way to solve a system of equations is by graphing each equation using a **T-chart**—like those that you filled in for the Lesson 7 practice. Graphing a system of linear equations provides a picture of the intersection. Follow these steps to solve a system of equations by graphing.

1. Set up an x and y T-chart for each equation. Find two ordered pairs for each equation: use $x = 0$ and find y and then use $y = 0$ and find x .
2. Graph both lines, using the ordered pairs that you generated in step 1.
3. Find the point of intersection and express it in the form (x, y) .

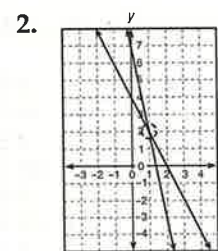
Example 1: Graph the equations $6x + 3y = 12$ and $5x + y = 7$ to find the solution.

1. $6x + 3y = 12$

x	y
0	4
2	0

$5x + y = 7$

x	y
0	7
$\frac{7}{5}$	0



3. The point of intersection—that is, the solution—is $(1, 2)$.

Substitute to Solve the System of Equations

You can also use **substitution** to solve a system of linear equations. Follow these steps to solve a system of linear equations by substitution.

Example 2: Solve the equations $6x + 3y = 12$ and $5x + y = 7$ by substitution.

1. Solve the first equation so that y is expressed in terms of x .

$$\begin{aligned} 6x + 3y &= 12 \\ 3y &= 12 - 6x \\ y &= 4 - 2x \end{aligned}$$

2. Substitute that value of y into the second equation and solve for x .

$$\begin{aligned} 5x + (4 - 2x) &= 7 \\ 3x + 4 &= 7 \\ 3x &= 3 \\ x &= 1 \end{aligned}$$

3. Substitute that value of x into the first equation and solve for y .

$$6(1) + 3y = 12$$

The solution is $(1, 2)$.

$$\begin{aligned} 6 + 3y &= 12 \\ 3y &= 6 \\ y &= 2 \end{aligned}$$

Key Ideas

- There are several ways to solve a system of linear equations.
- One method is to graph the equations to see the point of intersection.
- Another method is to find one value (x or y) and substitute it into the other equation to find the solution.

GED® TEST TIP

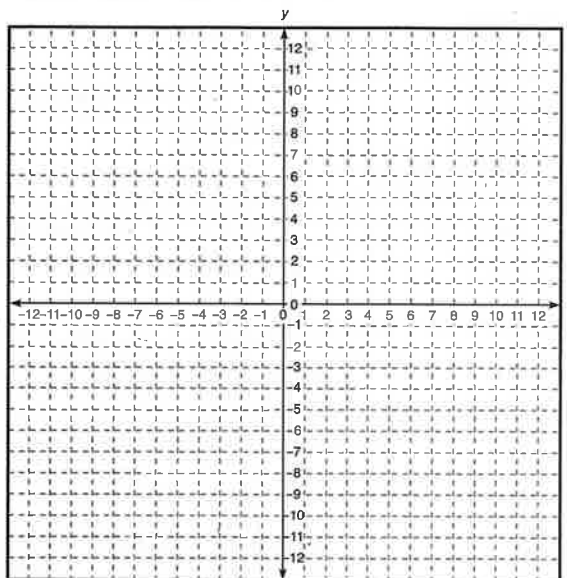
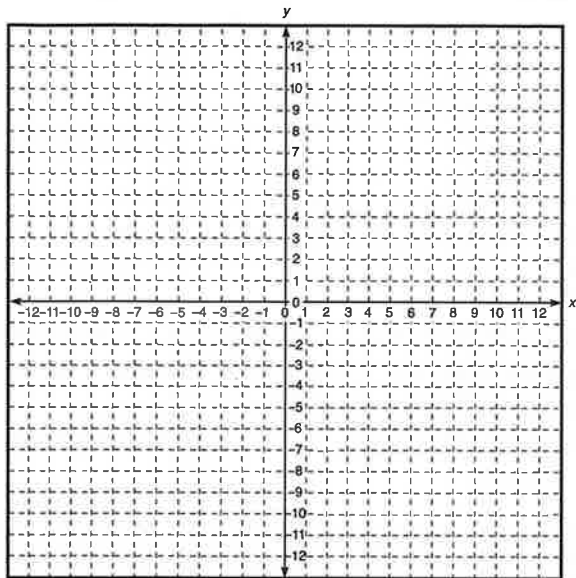
You may be asked whether a point is a solution to a system of equations, and you can use either of the methods shown here to find the answer.

Practice 10

A. Find two pairs of coordinates for each equation by making a T-chart. Use the coordinates to graph the lines and find the solution.

1. $y = 3x - 15$
 $x + y = 13$

2. $4x + 2y = 10$
 $y = -5x - 4$



B. Find the solution for the two equations by substitution. Express as an ordered pair in the form (x, y) .

3. $7x - y = 22$
 $4x + 2y = 10$

4. $x + y = 9$
 $2x - 3y = 8$

5. $y = 3x + 15$
 $5x - 2y = -26$

6. $10x - y = -1$
 $y = 12x$

C. Choose the one best answer for each question.

7. Where does the line with the equation $x - 2y = 4$ intersect with the line with the equation $6y + 5x = 4$?

- A. $(2, -1)$
- B. $(3, -2)$
- C. $(-2, 1)$
- D. $(-1, 2)$

8. Where does the line with the equation $y = x$ intersect with the line with the equation $y = -x$?

- A. $(1, -1)$
- B. $(-1, 1)$
- C. $(0, 0)$
- D. $(1, 0)$

9. Where does the line with the equation $y = -2$ intersect with the line with the equation $3y = 2x + 3$?

- A. $(-\frac{2}{9}, -2)$
- B. $(-\frac{1}{3}, -2)$
- C. $(-\frac{9}{2}, -2)$
- D. $(\frac{9}{2}, 2)$

10. Which of the following is the equation of a line that intersects $y = 4x + 2$?

- A. $2y = 8x + 2$
- B. $y = 4x - 2$
- C. $y = -4x + 2$
- D. $\frac{1}{2}y = 2x - 7$

Answers and explanations start on page 679.

Lesson 11: Patterns and Functions

Key Ideas

- A pattern is a sequence of numbers determined by a mathematical rule.
- A function is a rule that shows how one set of numbers is related to another set of numbers.
- To use a function, substitute values for variables and solve.

A **pattern** is a series of numbers or objects whose sequence is determined by a particular rule. You can figure out what rule has been used by studying the terms you are given. Think: What operation or sequence of operations will always result in the next term in the series? Once you know the rule, you can continue the pattern.

Example 1: Find the seventh term in the sequence: 1, 2, 4, 8, 16, . . .

1. Determine the rule. Each number in the sequence is two times the number before it.
2. Apply the rule. You have been given five terms and must find the seventh. Continue the pattern. The sixth term is $16 \times 2 = 32$, and the seventh term is $32 \times 2 = 64$.

A **function** is a rule that shows how the terms in one sequence of numbers are related to the terms in another sequence. Each distinct number entered into the function produces a unique output. For example, a sidewalk vendor charges \$1.50 for a slice of pizza. The chart below shows how much it would cost to buy one to six slices.

Number of Pizza Slices	1	2	3	4	5	6
Cost	\$1.50	\$3.00	\$4.50	\$6.00	\$7.50	\$9.00

Each number in the first row corresponds to a price in the second row. We could say that the amount a customer will pay is a function of (or depends upon) the number of slices the customer orders. This function could be written:

Cost = number of slices $\times$ \$1.50, or $C = n(\$1.50)$.

If you know the function and a number in the first set of numbers, you can solve for its corresponding number in the second set.

Example 2: Using the function $y = 3x + 5$, what is the value of y when $x = -3$?

1. Substitute the given value of x .
 $y = 3(-3) + 5$
2. Solve for y .
 $y = -9 + 5$
 $y = -4$

Example 3: Using the function $n = 100 - 4(3 + m)$, what is the value of n when $m = 6$?

1. Substitute the given value of m .
 $n = 100 - 4(3 + 6)$
2. Solve for n .
 $n = 100 - 4(9)$
 $n = 100 - 36$
 $n = 64$

GED® TEST TIP

To figure out what rule has been used to form a pattern, begin by finding the difference between each term and the term that follows it in the sequence.

Practice 11

A. Solve.

- Which number should come next in the following pattern?
-12, -9, -6, -3, _____
- What is the next number in the sequence?
21, 26, 31, 36, _____
- In the function $y = 4x + 10$, if $x = -2$, what is the value of y ?
- In the function $y = 2x(4 + x) - 2$, if $x = 3$, what is the value of y ?
- Each term in the second row is determined by the function $y = 2x - 1$.

x	1	2	3	4	5	...	12
y	1	3	5	7	9	...	

What number belongs in the shaded box?

- In the function $y = \frac{x+3}{6} - 8$, if $x = 21$, what is the value of y ?
- What is the next term in the pattern below?
1000, 500, 250, 125, 62.5, _____
- What is the next number in the sequence?
3, -5, 7, -9, 11, _____
- Each term in the second row is determined by the function $y = 3x + 5$.

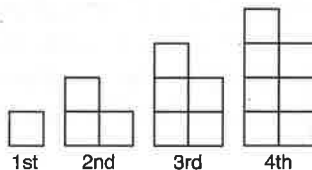
x	-2	-1	0	1	2	...	9
y	-1	2	5	8	11	...	

What number belongs in the shaded box?

- In the function $y = (x - 7) + 12$, if $x = -10$, what is the value of y ?

B. Choose the one best answer to each question.

Question 11 refers to the following drawing.



- How many blocks would be needed to build the 25th construction in the sequence?
A. 47
B. 49
C. 51
D. 55

- What is the sixth term in the sequence below?

-14, -8, -2, 4, ...

- 10
- 14
- 16
- 22

- The price per scarf is a function of the number of scarves purchased. The table shows the price per scarf for purchases of up to four scarves.

Number (n) of Scarves	1	2	3	4
Cost (c) per Scarf	\$5.00	\$4.75	\$4.50	\$4.25

Which of the following functions was used to determine the prices shown in the table?

- $c = n(\$5.00 - \$0.25)$
- $c = \$5.00 - \$0.25(n - 1)$
- $c = \$5.00 - \$0.25n$
- $c = \$5.00n - \$0.25n$

- Which of the following sequences of values of y could be created using the function $y = 4x - 3$?

- 1, 4, 7, 10, 13, ...
- 1, 5, 9, 13, 17, ...
- 1, 4, 8, 13, 19, ...
- 1, -1, -3, -5, -7, ...

Answers and explanations begin on page 680.

Lesson 12: Function Applications

Key Ideas

- Functions are used to make many common work calculations.
- FUNCTIONS CAN BE USED TO MAKE COMPARISONS.
- TO USE A FUNCTION, YOU MUST KNOW THE MEANING OF THE VARIABLES. THIS INFORMATION SHOULD BE GIVEN IN THE TEXT OF THE PROBLEM.

Functions are used in many business applications. For instance, they can be used to calculate profit, cost, employee wages, and taxes. On the *Mathematical Reasoning Test*, you will read about common work and life situations. The problems may contain or describe a function that you can use to solve the problem.

Example 1: Celino Advertising is finishing a series of print ads for a client. Finishing the project will cost \$2000 per day for the first seven days and \$3500 per day after seven days. The finishing costs for a project taking longer than 7 days can be found using the function $C = \$2000d + \$1500(d - 7)$, where C = the cost of finishing the project and d = the number of days. If the project takes 12 days to complete, what will the project cost?

$$\begin{aligned}\text{Use the function to solve the problem. } C &= \$2000d + \$1500(d - 7) \\ &= \$2000(12) + \$1500(12 - 7) \\ &= \$24,000 + \$1500(5) \\ &= \$24,000 + \$7500 \\ &= \$31,500\end{aligned}$$

You may be asked to use functions to make comparisons.

Example 2: Nita decides to join a health club. She gets brochures from two health clubs and compares the plans. Healthstars Fitness charges a one-time membership fee of \$250 and \$8 per month. Freedom Health Center charges \$25 per month. At both health clubs, the price (P) Nita will pay is a function of the number of months (m) she attends the club. The functions are as follows:

$$\text{Healthstars Fitness} \quad P = \$250 + \$8m$$

$$\text{Freedom Health Center} \quad P = \$25m$$

Nita plans to move in 18 months. If she attends a health club until she moves, which one offers the better price?

1. Find the price at Healthstars Fitness:
$$\begin{aligned}P &= \$250 + \$8m \\ &= \$250 + \$8(18) \\ &= \$250 + \$144 \\ &= \$394\end{aligned}$$
2. Find the price at Freedom Health Center:
$$\begin{aligned}P &= \$25m \\ &= \$25(18) \\ &= \$450\end{aligned}$$
3. Compare the results. Even though Nita will have to pay a large amount up front, **Healthstars Fitness** offers the better price.

GED® TEST TIP

There is often more than one way to work a problem, even when a function is given. Solve using the function. If you see another way to solve the problem, use it to check your answer.

Practice 12

A. Solve. You MAY use a calculator.

- The Chimney Sweep charges \$25 for a chimney inspection. If the customer purchases additional services, \$15 of the inspection fee is deducted. Let s = the cost of any additional services. The total cost (C) of an inspection and services can be determined by the function $C = \$25 + (s - \$15)$ where s is not 0.
 - Jan has her chimney inspected and purchases a smoke guard for \$89. How much will she be charged?
 - After an inspection, Ahmed decides to have a new damper installed for \$255. How much will he pay?
- Ricardo does a great deal of driving for his work. He generally estimates his driving time in hours (t) using the function $t = \frac{m}{60}$, where m = the number of miles.
 - How many hours will it take Ricardo to drive 330 miles?
 - How many hours will it take Ricardo to drive 255 miles?

B. Choose the one best answer to each question. You MAY use a calculator.

Questions 4 and 5 refer to the following information.

Alicia is considering three job opportunities. At all three jobs, weekly pay (P) is a function of the number of hours (h) worked during the week. The functions are shown below:

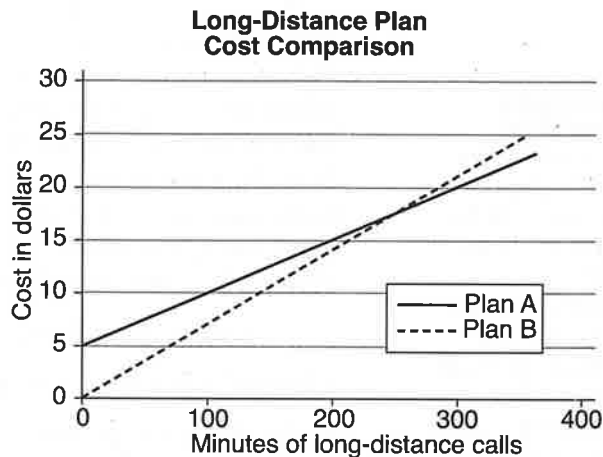
Job 1 $P = \$9.75h$

Job 2 $P = \$70 + \$8.40h$

Job 3 $P = \$380 \times \frac{h}{38}$

- If Alicia works 30 hours in a week, how much more will she earn at Job 2 than at Job 1?
 - \$5.33
 - \$29.50
 - \$40.50
 - \$59.00

- A customer's phone charges are a function of the number of minutes of long-distance calls made. The graph shows a comparison of two plans available.



- Michelle looks at her previous phone bills and finds that she makes about 350 minutes of long-distance calls per month. Which plan is better for her?
- Craig usually makes about 150 minutes of long-distance calls per month. Which plan is better for him?

- If Alicia works 40 hours per week, which of the following is a true statement?
 - Alicia will earn the least at Job 3.
 - Job 1 will pay more than Job 3.
 - Job 3 will pay more than Job 2.
 - Alicia will earn the most at Job 2.
- A company is awarded a \$95,000 job that will cost \$5,400 per day in expenses. Profits (P) can be calculated using $P = \$95,000 - \$5,400d$, where d = days. What is the company's profit if the job takes 14 days to complete?
 - \$10,800
 - \$19,400
 - \$66,100
 - \$75,600

Answers and explanations begin on page 680.

Lesson 13: Function Notation

Functions can be written with **function notation**. Take the function in Example 1 of the previous lesson: $C = \$2000d + \$1500(d - 7)$, where C = the cost of finishing a project and d = the number of days the project takes. In function notation, the cost would be written as $f(d)$ instead of C , so the function is written as follows: $f(d) = \$2000d + \$1500(d - 7)$. The notation $f(d)$ is interpreted as “the function of d ” or “ f of d .”

Key Ideas

- In function notation, the result of the function is expressed in terms of the variable in the function, for example, $f(x) = x + 4$.
- The domain of a function is the set of valid input values, or those values of the variable for which the function is defined.
- The range of a function is the set of output values that it can generate, given the values in its domain.

The **domain** of a function is the set of values for which the function is defined or makes sense. In the above example, the domain is all the values that d can take. Since d is an amount of time, it cannot be negative. Furthermore, in this case if $d < 7$, then the function begins to subtract from the \$2000 a day cost for finishing the project, which does not make sense (as explained in the example, \$2000/day is the rate for the first seven days, and additional days cost even more; no days cost less than \$2000). Thus, the domain of $f(d)$ is $d \geq 7$.

Take a look at the following example involving a quadratic expression.

Example 1: What is the domain of $f(x) = \frac{1}{1-x^2}$?

1. The domain is the set of values that x can take on to produce a defined value for $f(x)$. Division by zero is undefined.

2. Calculate the values of x that produce a denominator of zero.

$$\begin{aligned}1 - x^2 &= 0 \\(1 + x)(1 - x) &= 0 \\x &= \pm 1\end{aligned}$$

3. The domain is all real numbers not equal to -1 or 1 .

The **range** of a function is the set of outputs or results of the function. Look again at the example $f(d) = \$2000d + \$1500(d - 7)$. When d is as small as possible—that is, when $d = 7$ —then $f(d) = \$2000 \times 7 + \$1500(7 - 7) = \$14,000$. There is no upper limit to how many days a job could take, so there is no upper limit to the cost or to $f(d)$. Thus, the range of $f(d)$ is numbers greater than or equal to 14,000.

Example 2: What is the range of $f(x) = x^2$?

1. First, determine the domain of this function. Any number can be squared, so the domain is all values of x .

2. Find the minimum value of $f(x)$. A squared real number is never negative. When $x = 0$, $f(x) = 0$ and that is the function’s minimum value.

3. Find the maximum value of $f(x)$. As x moves away from zero, whether in a positive or negative direction, $f(x)$ takes on a greater positive value. There is no limit on how large $f(x)$ can be. Thus, the range of $f(x)$ is numbers greater than or equal to zero.

Note that sometimes graphing a function is very helpful to find its domain and range.

GED® TEST TIP

Sometimes you may be given a function and an input value and be asked to solve for the output, and sometimes you may be given the output and be asked to solve for the input. In either case, the question will always provide enough information to figure out what the function is.

A question could give you the value of a function and ask what value(s) of the variable would produce that result.

Example 3: Rafael enjoys baking cookies with his nieces and nephews. For every cup of oats, they can make 24 oatmeal cookies. If they made 180 oatmeal cookies today, how many cups of oats did they use?

1. Write the function.

$f(c) = 24c$, where c is the number of cups of oats and $f(c)$ is the number of cookies.

2. Substitute the value of the outcome for $f(c)$.

$$180 = 24c$$

3. Solve for c .

$$c = \frac{180}{24} = 7.5$$

Rafael and his family used 7.5 cups of oats to make the cookies.

Practice 13

A. Find the domain and range of the following functions.

1. $f(x) = x - 2$

2. $f(x) = \frac{5}{x+3}$

3. $f(x) = x^2 + 4x + 4$

4. $f(b) = -|-6b|$

5. $f(x) = -(x^2 - 1)$

6. $f(z) = \frac{z^2 - 9}{z^2 - 6z + 9}$

B. Solve.

7. Carla's daughter is participating in a summer reading program at the library. She is reading two books a week. Which of the following functions expresses her summer reading, where w is the number of weeks in the program?

A. $f(w) = w + 2$

B. $f(w) = 2w$

C. $f(w) = 2b$

D. $2f(w) = w$

8. Bryan is on a road trip with his best friend, and they like to get up early. Each day, they drive 100 miles before breakfast, and then they drive an average of 60 miles per hour for each hour they are on the road until they stop for the day. They never spend more than 8 hours on the road after breakfast. If h is the number of hours they drive after breakfast, which of the following are the domain and range of the function that represents their daily driving?

A. Domain is $0 \leq h \leq 8$; range is $100 \leq f(h) \leq 580$.

B. Domain is $100 \leq h \leq 580$; range is $0 \leq f(h) \leq 8$.

C. Domain is $h \geq 0$; range is $f(h) \geq 100$.

D. Domain is $h \leq 8$; range is $60 \leq f(h) \leq 100$.

Answers and explanations begin on page 681.



Equations, Inequalities, and Functions Practice Questions

Directions: You MAY use your calculator.

1. Which of the following expressions is equal to $6 - 4(x + 3)$?

A. $4x + 3$
B. $4x - 9$
C. $-4x - 3$
D. $-4x - 6$

2. Three increased by the product of 4 and a number is equal to the same number decreased by 6. What is the number?

A. -3
B. -1
C. 1
D. 3

3. What is the equation of a line with a slope of -4 that passes through the point $(1, 2)$?

A. $y = -4x + 2$
B. $y = -4x + 6$
C. $y = 4x + 1$
D. $y = 4x + 6$

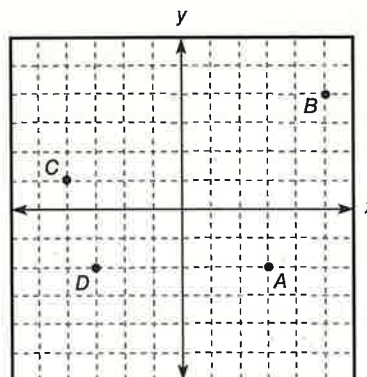
4. The ordered pair $(-2, -1)$ is a solution to which of the following equations?

A. $-4x - y = 7$
B. $4x + y = -7$
C. $4x - y = -7$
D. $-4x + y = -7$

5. For a two-week period, Jan earned \$150 less than twice Tom's earnings. Together Jan and Tom earned \$1380. How much did Tom earn?

A. \$720
B. \$660
C. \$510
D. \$360

Questions 6 and 7 refer to the following graph.



6. If the graph of the equation $y = -x - 5$ were drawn on the grid, which of the points would be on the line?

A. A
B. B
C. C
D. D

7. If a line were drawn through points B and C, what would be the slope of the line?

A. -3
B. $-\frac{1}{3}$
C. $\frac{1}{3}$
D. 3

8. Samuel is paid \$350 per month plus a 10 percent commission on his total sales for the month. If he needs to earn at least \$2100 per month, which of the following expressions represents the total sales for the month (s) Samuel needs to achieve?

A. $s \geq \$17,500$
B. $s \leq \$17,500$
C. $s \leq \$21,000$
D. $s \geq \$21,000$

9. For its Checking Plus account, a bank charges \$3.95 per month plus \$0.10 for each check written after the first ten checks written that month. The function to find the total monthly fee (F) is $F = \$3.95 + \$0.10(n - 10)$, where n = number of checks written.

Greg writes 24 checks in March. How much will he pay in fees for the month?

- A. \$6.35
B. \$5.35
C. \$2.40
D. \$1.40
10. The sum of four consecutive odd numbers is 104. What is the largest number?

- A. 23
B. 25
C. 27
D. 29

11. Which value for x makes the inequality $x > 400$ true?

- A. 7^3
B. 4^4
C. 3^6
D. 6^3

12. Four less than the product of a number (x) and 5 is equal to 8 more than the sum of 2 and the product of 3 and the number. Which of these equations could be used to find the value of x ?

- A. $4 - 5x = 8 + 2 + 3x$
B. $5x - 4 = 8 + 2 + 3x$
C. $5x = 4 - 8 + 2 + 3x$
D. $5x - 4 = 8 + 2 + 3$

13. What is the slope of a line that passes through the points (2, 4) and (4, 6)?

- A. -2
B. 1
C. 2
D. 4

14. What is the next number in the sequence?

1, 7, 14, 22, ...

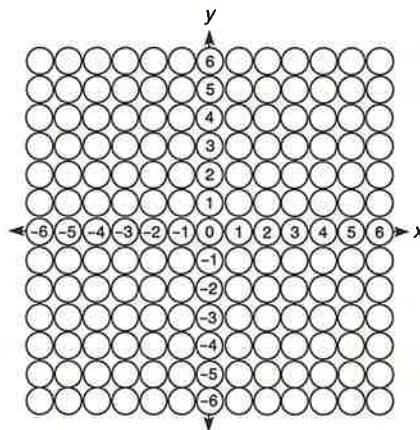
- A. 30
B. 31
C. 32
D. 33

15. What is the y -intercept of the line with a slope of 2 that passes through the point (1, 2)?

- A. -2
B. $-\frac{1}{2}$
C. 0
D. $\frac{1}{2}$

16. The graphs of the equations $y = x + 3$ and $y = -2x - 3$ are drawn on a coordinate grid.

In the coordinate plane below, darken the circle corresponding to the point at which the two lines intersect.



17. Which of the following shows the product of -7 and x decreased by the sum of 8 and y ?

- A. $-7x - (8 + y)$
- B. $-7x - 8 + y$
- C. $(8 + y) - 7x$
- D. $(7 + x) - (8 + y)$

18. What are the possible solutions for the quadratic equation $x^2 - 5x = 24$?

- A. -8 and 3
- B. -6 and 4
- C. 8 and -3
- D. 4 and -6

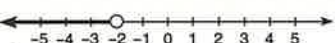
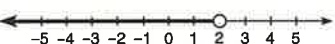
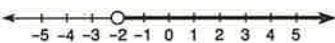
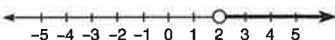
19. Cynthia is 6 times as old as Rebecca. In 6 years, Cynthia will be only 3 times as old as Rebecca. How old is Rebecca now?

- A. 3
- B. 4
- C. 8
- D. 12

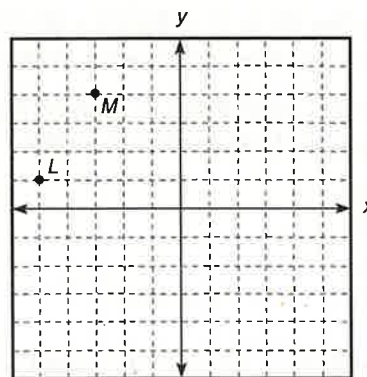
20. The graph of which equation will pass through points $(0, -3)$ and $(5, 7)$?

- A. $x = \frac{1}{2}x - 3$
- B. $y = 2x - 3$
- C. $y = 2x + 7$
- D. $y = -2x + 3$

21. Which of the following graphs represents the solution set of the inequality $-2(x - 6) > 8$?

- A. 
- B. 
- C. 
- D. 

Questions 22 and 23 refer to the following graph.



22. Which of the following ordered pairs shows the location of point L ?

- A. $(-5, -1)$
- B. $(-5, 1)$
- C. $(-1, 5)$
- D. $(1, -5)$

23. What is the slope of the line that passes through points L and M ?

- A. $-\frac{3}{2}$
- B. $-\frac{2}{3}$
- C. $\frac{2}{3}$
- D. $\frac{3}{2}$

24. Bob, Celia, Sam, and Daniel contributed money to buy their boss a retirement gift. Sam and Daniel each gave the same amount of money. Celia gave \$12 more than Daniel. Bob gave half as much as Celia gave. If the four workers gave a total of \$81, how much did Sam give?

- A. \$9
- B. \$15
- C. \$18
- D. \$21

25. What is the solution set of the inequality $-7x - 4 \geq x - 28$

A. $x \leq 3$
 B. $x \geq 3$
 C. $x \leq -3$
 D. $x \geq -4$

26. A line goes through the points (0, 2) and (1, 3). What is the equation of the line?

A. $y = -\frac{1}{2}x + 3$
 B. $y = \frac{1}{2}x + 2$
 C. $y = x + 2$
 D. $y = x + 3$

27. In a recent election, Perez got 5512 more votes than $\frac{1}{3}$ of the leading candidate's votes. Together the two candidates received 18,072 votes. How many people voted for Perez?

A. 6,024
 B. 8,652
 C. 9,420
 D. 12,560

28. What is the value of x in the equation $-4(x + 2) - 10 = 5x$?

A. $x = -18$
 B. $x = -2$
 C. $x = 2$
 D. $x = 18$

29. Marcia counts the \$5 bills and \$10 bills in her cash register drawer. She counts a total of 35 bills with a total value of \$240. If x = the number of \$5 bills in the drawer, which of the following equations could be used to find the number of \$5 bills in the drawer?

A. $\$5x + \$10x = 35$
 B. $\$5x + \$10x + 35 = \$240$
 C. $\$5(35 - x) + \$10x = \$240$
 D. $\$5x + \$10(35 - x) = \$240$

30. The sum of five consecutive numbers is 370. What is the fourth number in the sequence?

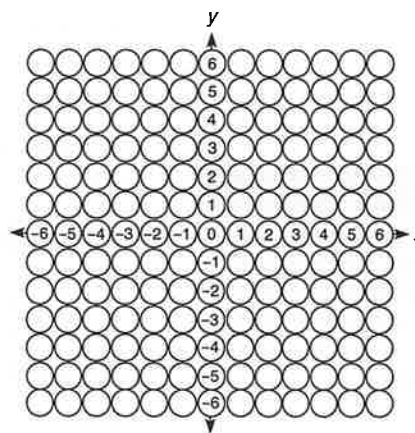
A. 73
 B. 74
 C. 75
 D. 76

31. A baseball pitcher's earned run average (E) is a function of the number of earned runs (r) given up and innings pitched (i). The function is written $E = \frac{9r}{i}$. What is the earned run average of a pitcher who gives up 8 runs in 18 innings?

Write your answer on the line below.

32. Point B is located halfway between $(-1, 5)$ and $(-1, -3)$. Name the location of point B .

In the coordinate plane below, darken the circle corresponding to the location of point B .



Answers and explanations begin on page 681.

Geometry

Lesson 1: Plane Figures

Key Ideas

- Plane figures are classified by the properties of their sides and angles.
- The sum of the interior angles of any four-sided plane figure is 360° .
- By using the properties of any four-sided plane figure and algebraic reasoning, you can find missing angle measures.

GED® TEST TIP

Don't rely on sight alone to identify a geometric figure. Read carefully to see which properties are given. Then identify the figure based on the properties.

Four-Sided Plane Figures

A **plane figure** is a set of line segments, all lying on a single plane. To prepare for the GED® *Mathematical Reasoning Test*, learn the properties of each shape. You will need to identify the characteristics of different types of four-sided plane figures and draw conclusions about their angles and sides.

You are already familiar with rectangles and squares. A **rectangle** is a four-sided figure with four right angles. The opposite sides (sides across from each other) are the same length, and they are parallel.

A **square** is actually a kind of rectangle. It, too, has four right angles with parallel opposite sides. However, a square has one additional property: its four sides are all the same length.

A **parallelogram** is a four-sided figure whose opposite sides are parallel and the same length. In addition, its opposite angles (the angles diagonally across from each other) are also equal in measure. A special parallelogram, called a **rhombus** (not shown), has four sides of equal length.

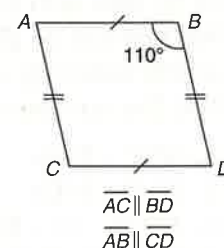
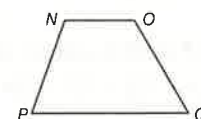
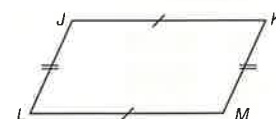
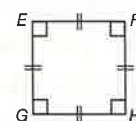
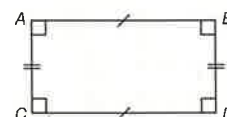
A **trapezoid** is a four-sided figure with exactly one pair of parallel sides. The definition of a trapezoid does not dictate the measure of the angles or the lengths of the sides.

All four-sided plane figures have one important property in common. The sum of the measures of the interior angles is 360° . You can use this fact to find a missing angle measure.

Example: In figure $ABCD$, the opposite sides are parallel. What is the measure of $\angle A$?

- Identify the figure. The notation on the drawing tells you that the opposite sides are equal in measure. Since they are also parallel, the figure is a parallelogram.
- Find the measure of $\angle C$. The opposite angles of a parallelogram are equal in measure; therefore, $m\angle C = m\angle B$. Both $\angle B$ and $\angle C$ measure 110° .

Sides with the same markings are equal.



3. Find the measure of $\angle A$. You know the measures of $\angle A$ and $\angle D$ are equal and that the sum of all four angles equals 360° . Let $x = m\angle A$. Therefore, $2x =$ the sum of $m\angle A$ and $m\angle D$. Write an equation and solve.
- $$\begin{aligned} 2x + 110^\circ + 110^\circ &= 360^\circ \\ 2x + 220^\circ &= 360^\circ \\ 2x &= 140^\circ \\ x &= 70^\circ \end{aligned}$$

The measure of $\angle A$ is 70° .

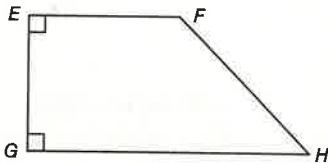
Practice 1

- A. List the names of four-sided plane figures introduced on page 388 that can exhibit the following properties. Write *None* if no four-sided plane figure has the given property.

- | | |
|---------------------------------------|--|
| 1. four right angles | 6. opposite angles are equal in measure |
| 2. opposite sides are equal in length | 7. all four sides are equal in length |
| 3. exactly one pair of parallel sides | 8. sum of interior angles is 360° |
| 4. all angles are equal in measure | 9. sides are all of different lengths |
| 5. only three right angles | 10. four equal angles and four equal sides |

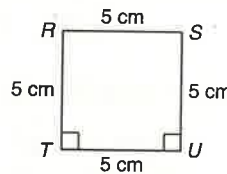
- B. Choose the one best answer to each question.

Questions 11 and 12 refer to the following figure.



11. Angle F is 20° more than three times the measure of $\angle H$. What is the measure of $\angle F$?
- A. 40°
 B. 120°
 C. 140°
 D. 180°
12. In order for figure $EFGH$ to be a trapezoid, which of the following must be a true statement?
- A. EF is the same length as FH .
 B. $EF \parallel GH$.
 C. $m\angle F = m\angle H$.
 D. $m\angle G = m\angle F$.

Question 13 refers to the following figure.



13. If the opposite sides in figure $RSUT$ are parallel, what is the measure of $\angle R$?
- A. 5°
 B. 20°
 C. 90°
 D. 270°
-
14. A four-sided plane figure has sides measuring 10, 15, 10, and 15. The opposite angles are equal, but there are no right angles. What is the figure?
- A. rhombus
 B. rectangle
 C. parallelogram
 D. trapezoid

Answers and explanations begin on page 682.

Lesson 2: Triangles

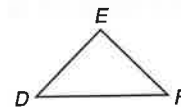
Key Ideas

- Triangles are named by their vertices.
- They are classified in two ways: by their side lengths and by their angle measures.
- The sum of the interior angles of any triangle is 180° .

The Properties of Triangles

A **triangle** is a closed three-sided plane figure. From the definition, we can infer other properties. Since a triangle has three sides, it must also have three interior angles and three vertices.

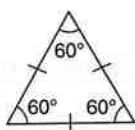
A triangle is named by writing its vertices in any order. The triangle shown at right could be named $\triangle DEF$. Its sides are DE , EF , and DF .



Triangles can be classified by the lengths of their sides and by the measures of their angles. In the figures below, sides with the same number of marks are equal.

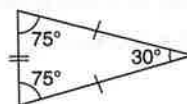
equilateral triangle

All sides are equal in length. Note that the angles also are equal.



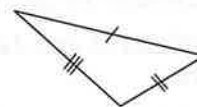
isosceles triangle

Exactly two sides are equal in length. Note that the two angles opposite these sides are equal.



scalene triangle

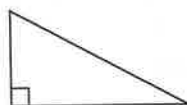
No sides are equal in length, and no angles are equal.



Classified by Angle Measures

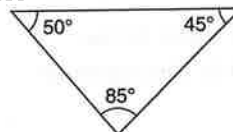
right triangle

One angle measures 90° .



acute triangle

All angles measure less than 90° .



obtuse triangle

One angle is greater than 90° .

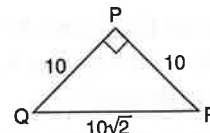


Each triangle can be classified in two ways.

Example 1: What kind of triangle is $\triangle PQR$?

1. Classify by its sides: Two sides have the same length, so $\triangle PQR$ is an isosceles triangle.
2. Classify by its angles: $\angle P$ is a right angle, so $\triangle PQR$ is a right triangle.

$\triangle PQR$ is a **right isosceles triangle**.



GED® TEST TIP

Classify triangles by their properties, not by how they look. For example, the triangle in Example 1 may not immediately look like a right triangle because the right angle is at the top.

The sum of the measures of the interior angles of any triangle is 180° . We can use this fact to solve for a missing angle.

Example 2: In $\triangle ABC$, $\angle A$ measures 55° and $\angle B$ measures 100° . What is the measure of $\angle C$?

Write an equation and solve. $55^\circ + 100^\circ + \angle C = 180^\circ$

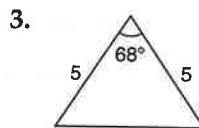
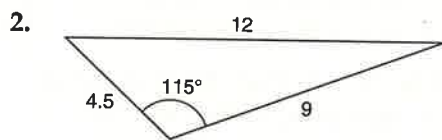
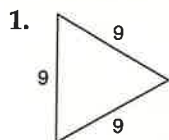
$$155^\circ + \angle C = 180^\circ$$

$$\angle C = 25^\circ$$

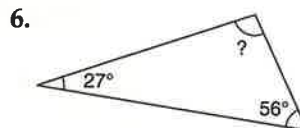
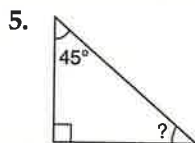
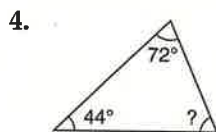
The measure of $\angle C$ is 25° .

Practice 2

A. Classify each triangle in two ways.

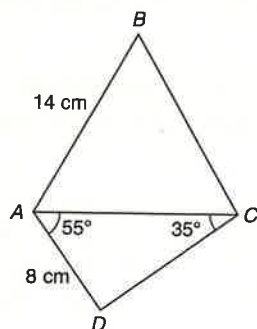


B. Find the measure of the unknown angle in each triangle.



C. Choose the one best answer to each question.

Questions 7 and 8 refer to the following figure.



7. If $\angle DAB$ measures 115° and $\angle DCB$ measures 95° , what is the length of side AC in centimeters? (Hint: Use the facts in the problem to find $m\angle BAC$ and $m\angle BCA$.)

- A. 6
- B. 8
- C. 14
- D. 22

8. What kind of triangle is $\triangle ACD$?

- A. isosceles
- B. acute
- C. right
- D. obtuse

9. One angle in a scalene triangle measures 38° , and another angle measures 56° . What is the measure of the third angle?

- A. 38°
- B. 56°
- C. 86°
- D. 124°

Answers and explanations begin on page 682.

Lesson 3: Pythagorean Relationship

Key Ideas

- The hypotenuse is the longest side of a right triangle. It is found opposite the right angle.
- The sum of the squares of the legs of a right triangle equals the square of the hypotenuse.

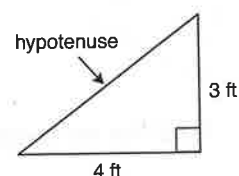
As you know, a right triangle has one right angle. The side directly across from the right angle, called the **hypotenuse**, is the longest side of the right triangle. The remaining sides, the line segments forming the right angle, are the **legs** of the triangle.

Thousands of years ago, people found a special relationship, called the **Pythagorean relationship**, among the sides of a right triangle. You can use this relationship to find the measure of any side of a right triangle if the other two side measures are known.

Pythagorean relationship $a^2 + b^2 = c^2$; a and b are legs, and c is the hypotenuse of a right triangle

In other words, the square of the hypotenuse is equal to the sum of the squares of the two legs of the right triangle.

Example 1: What is the length of the hypotenuse of the right triangle shown in the diagram?



1. The lengths of the legs are 3 ft and 4 ft. Let one leg equal a and the other equal b .

2. Solve for c . Substitute the values.

3. When one side of an equation equals a squared variable, isolate the variable by finding the square root of both sides.

$$\begin{aligned} a^2 + b^2 &= c^2 \\ 3^2 + 4^2 &= c^2 \\ 9 + 16 &= c^2 \\ 25 &= c^2 \\ \sqrt{25} &= c \\ 5 &= c \end{aligned}$$

GED® TEST TIP

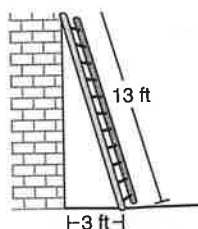
Watch for right triangles with sides in a 3-4-5 or 5-12-13 ratio. These triangles often appear on math tests because the measures of the sides are all whole numbers.

The length of the hypotenuse is 5 feet.

The Pythagorean relationship can also be used to solve for the length of a leg.

Example 2: If John places a 13-foot ladder 3 feet from the base of a wall, how far up the wall will the ladder reach to the nearest tenth of a foot?

The wall, ground, and ladder form a right triangle. The hypotenuse is 13 ft in length. One leg is 3 ft. You need to find the length of the other leg.



$$\begin{aligned} a^2 + b^2 &= c^2 \\ 3^2 + b^2 &= 13^2 \\ 9 + b^2 &= 169 \\ b^2 &= 160 \\ b &= \sqrt{160} \\ b &\approx 12.6 \end{aligned}$$

The ladder will extend **12.6 feet** up the wall.

NOTE: Most of the time, you will need to use your calculator for the final step when using the Pythagorean relationship. To find the square root of 160 on the TI-30XS MultiView™ calculator, press: $\boxed{2nd} \boxed{x^2} \boxed{160} \boxed{enter}$ (use $\boxed{\leftarrow} \boxed{\rightarrow}$ to convert the result from a radical to a decimal format).

Some right triangles display special proportions, which are worth memorizing.

- A right triangle whose angles are 45° , 45° , and 90° displays the following relationship:
leg : leg : hypotenuse = $x : x : x\sqrt{2}$.
- A right triangle whose angles are 30° , 60° , and 90° displays the following relationship:
leg : leg : hypotenuse = $x : x\sqrt{3} : 2x$.

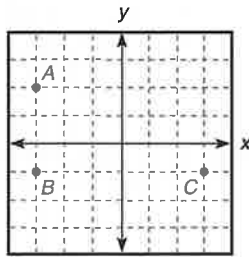
Practice 3

A. The lengths of two sides of a right triangle are given. Find the length of the remaining side to the nearest tenth unit. You MAY use a calculator.

- | | | |
|---|---|---|
| 1. leg a : 8 in.
leg b : 8 in.
hypotenuse c : ? in. | 4. leg a : ? m
leg b : 3 m
hypotenuse c : 6 m | 7. leg a : 7 cm
leg b : 10 cm
hypotenuse c : ? cm |
| 2. leg a : 9 yd
leg b : 12 yd
hypotenuse c : ? yd | 5. leg a : 6 mm
leg b : ? mm
hypotenuse c : 10 mm | 8. leg a : 15 in.
leg b : ? in.
hypotenuse c : 30 in. |
| 3. leg a : 1.5 cm
leg b : 2 cm
hypotenuse c : ? cm | 6. leg a : ? ft
leg b : 5 ft
hypotenuse c : 18 ft | 9. leg a : 4 km
leg b : 5 km
hypotenuse c : ? km |

B. Choose the one best answer to each question.

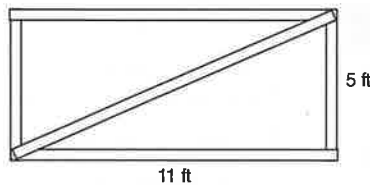
10. On a coordinate plane, points A , B , and C can be connected to form a right triangle.



What is the distance from A to C , to the nearest tenth of a unit? (*Hint: Count units to find the lengths of the sides and use the Pythagorean relationship to find the distance between the points.*)

- A. 5.2
B. 6.7
C. 8.4
D. 10.1
11. The two shorter sides of a right triangle measure 18 ft and 24 ft. What is the measure in feet of the third side?
- A. 25
B. 28
C. 30
D. 42

12. Jan has built a rectangular frame out of wood to use for the bottom of a platform. He wants to add a diagonal brace as shown in the drawing below.



What will the length of the brace be, to the nearest tenth of a foot?

- A. 16.0
B. 13.7
C. 12.8
D. 12.1
13. The hypotenuse of a right triangle measures 39 inches. If one leg measures 15 inches, what is the measure of the other leg, in inches?
- A. 36
B. 24
C. 18
D. 12

Answers and explanations begin on page 683.

Lesson 4: Perimeter and Area

Key Ideas

- Perimeter is the distance around a figure.
- Area is the measure of the space inside a flat figure. It is measured in square units.
- You need to memorize the formulas on this page except those for area of a parallelogram and area of a trapezoid.

Perimeter is the distance around a figure. To find perimeter, simply add the lengths of the sides. For common figures, you can apply a formula to find the perimeter. You need to memorize these formulas.

square	Perimeter = $4 \times \text{side}$	$P = 4s$
rectangle	Perimeter = $2 \times \text{length} + 2 \times \text{width}$	$P = 2l + 2w$
triangle	Perimeter = $\text{side}_1 + \text{side}_2 + \text{side}_3$	$P = a + b + c$

Example 1: A rectangle is 16 inches long and 9 inches wide. What is the perimeter of the rectangle?

$$\begin{aligned}
 \text{Use the formula: Perimeter} &= 2 \times \text{length} + 2 \times \text{width} \\
 &= 2 \times 16 + 2 \times 9 \\
 &= 32 + 18 \\
 &= \mathbf{50 \text{ in.}}
 \end{aligned}$$

Area is the measure of the space inside a flat figure. Area is measured in square units. For example, if the sides of a figure are measured in inches, its area will be measured in square inches. The formulas for finding area are shown below.

square	Area = side^2	$A = s^2$
rectangle	Area = $\text{length} \times \text{width}$	$A = lw$
parallelogram	Area = $\text{base} \times \text{height}$	$A = bh$
triangle	Area = $\frac{1}{2} \times \text{base} \times \text{height}$	$A = \frac{1}{2}bh$
trapezoid	Area = $\frac{1}{2} \times \text{height} \times (\text{base}_1 + \text{base}_2)$	$A = \frac{1}{2}h(b_1 + b_2)$

GED® TEST TIP

Several area formulas appear on the GED® formula sheet. You can check there if you don't remember a specific formula on test day. However, memorizing the most common ones (rectangle, triangle, and circle) will likely save you time as you're testing.

Three of the formulas mention two new measures: base and height. The **base** is one side of the figure. The **height** is the length from the vertex to the base, forming a right angle to the base.

Example 2: Find the area of figure ABCD.

1. Identify the figure. ABCD is a parallelogram.
2. Find the facts you need. To use the formula for finding the area of a parallelogram, you need to know the height and the length of the base. Ignore the length of side BD.

3. Use the formula $\text{Area} = \text{base} \times \text{height}$.

$$\begin{aligned}
 \text{Area} &= 12 \times 7 \\
 &= \mathbf{84 \text{ sq cm or } 84 \text{ cm}^2}
 \end{aligned}$$

